MHT-CET Weightage: 5%

Semiconductor Fundamentals

Semiconductor Properties:

- Semiconductors have conductivity between conductors and insulators.
- At low temperatures, their conductivity is low and with rise in temperature their conductivity increases.
- Pure semiconductors are called intrinsic semi conductor (Ex. Silicon and Germanium).

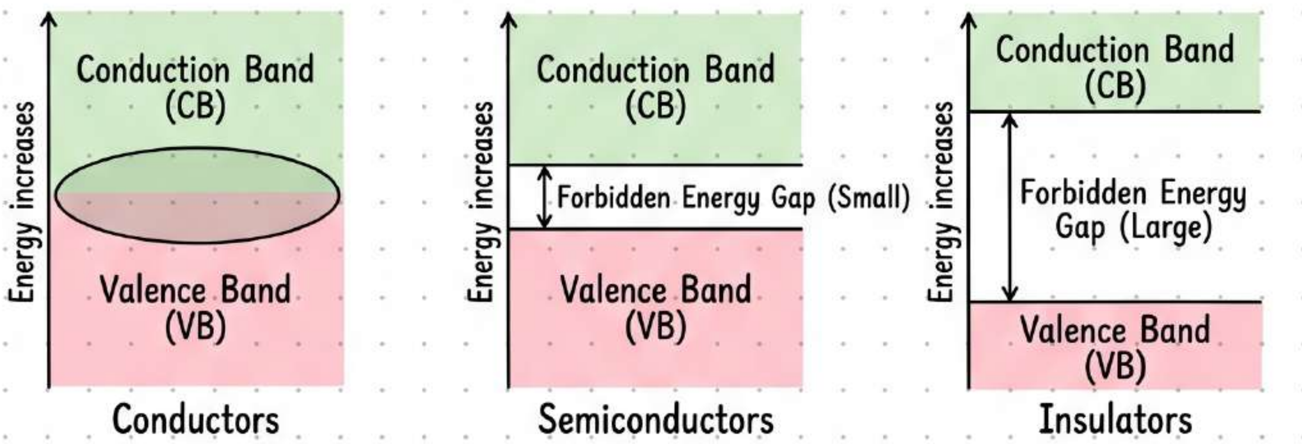
Energy Bands:

- In all solids, different energy levels combine to form two bands: valence band and conduction band.
  - Valence band (VB) contains valence electrons. It may be partially or fully filled.
  - Conduction band (CB) contains free electrons. It may be empty or partially filled.

Forbidden Energy Gap:

- The energy difference between valence band and conduction band is called forbidden energy gap. The forbidden energy gap is less for conductors and large for insulators and in between for semiconductors.

Energy Band Diagram



P-N Junction & Zener Diodes

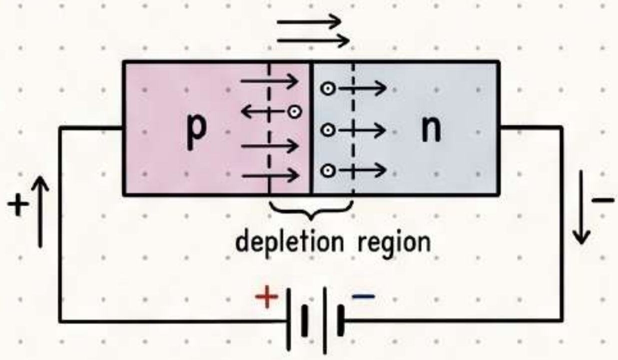
P-N Junction Diode:

A p-n junction is a single piece of semiconductor, one half of which is p-type and the other half is n-type. The region near the junction is called depletion layer.

Biasing Conditions:

Forward Bias:

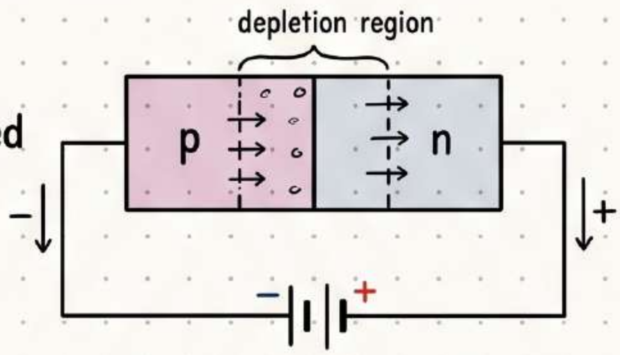
When p-type is connected to positive and n-type connected to negative terminal then it is forward biased.
In forward bias the diode offers low resistance and depletion region becomes narrower. It is similar to 'ON' in an electrical switch.



Forward Bias

Reverse Bias:

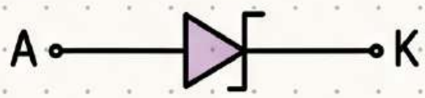
In reverse bias p-type is connected to negative terminal and n-type is connected to positive terminal.
In reverse bias the diode offers high resistance, does not conduct and depletion region becomes widened. It is similar to 'OFF' in an electrical switch.



Reverse Bias

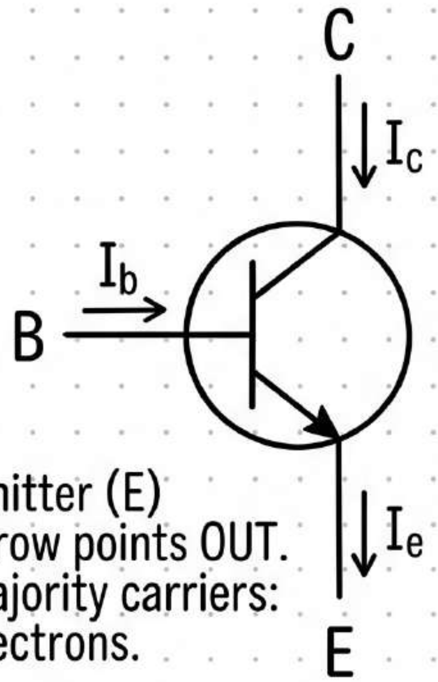
Zener Diode:

A properly and highly doped p-n junction diode which operates in reverse bias condition is called 'Zener diode'.
Silicon is preferred for making Zener diodes.
Zener diode is operated in reverse bias, which operates at a voltage called 'Zener voltage'.
Zener diode is used as a 'Voltage regulator'.



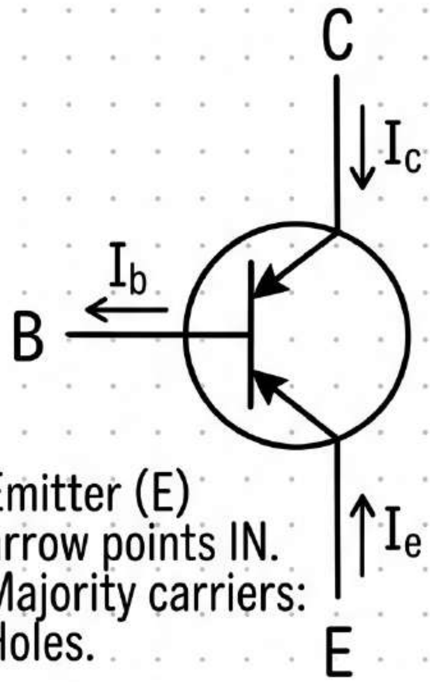
Transistors & Amplification

NPN Configuration



Emitter (E)
arrow points OUT.
Majority carriers:
Electrons.

PNP Configuration



Emitter (E)
arrow points IN.
Majority carriers:
Holes.

Transistor means transfer resistor. Works as an amplifier and switch.

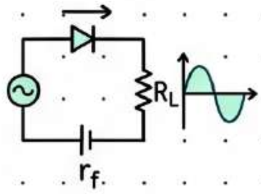
MHT-CET Weightage: 5%

Semiconductor Formulae

• RECTIFIER EFFICIENCIES

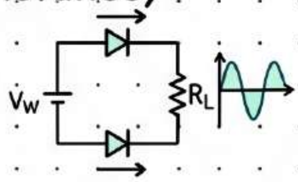
→ Rectifier efficiency (η) = $\dfrac{\text{dc output power}}{\text{ac input power}}$

→ Half wave rectifier efficiency (η) $\eta = \dfrac{0.406 \times R_L}{r_f + R_L}$



(where r_f = diode forward resistance, R_L = load resistance)

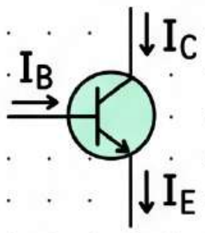
→ Full wave rectifier efficiency (η) $\eta = \dfrac{0.812 \times R_L}{r_f + R_L}$



• TRANSISTOR GAINS

→ Current gain $\beta = \dfrac{\Delta I_C}{\Delta I_B}$

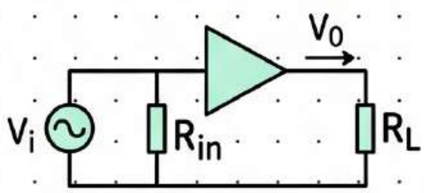
$\alpha = \dfrac{\Delta I_C}{\Delta I_E}$



→ Relation between α and β : $\beta = \dfrac{\alpha}{1 - \alpha}$

• AMPLIFICATION FACTORS

→ Amplification factor $A = \dfrac{V_0}{V_i}$



(where V_0 = Output voltage
V_i = Input voltage)

→ Voltage gain $A_v = \dfrac{\Delta V_{CE}}{\Delta V_{RE}} = \beta \times \dfrac{R_L}{R_{in}}$

(where ΔV_{CE} = Change in output voltage
ΔV_{RE} = Change in input voltage)

→ Power gain A_p = Current gain × voltage gain = $\beta^2 \times \dfrac{R_L}{R_{in}}$